

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/DEB_SASWI_dv")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3333, C2 = 6666, C3 = 1222)

text_priors <- '
Distrib (pAm,          TruncNormal_cv,          100,          0.2,          40, 2000);
Distrib (Fm,           TruncNormal,              0.1,           0.2,          0.01, 0.8);
Distrib (r_pAm_pM,     TruncNormal_cv,          0.8,           0.1,          0.1, 2);
Distrib (kap,          Uniform,                  0.1,           0.9);
Distrib (Ehp,          TruncNormal_cv,          100,           0.2,          40, 300);
Distrib (E_coc,        TruncNormal_cv,              30,           0.2,          1, 300);
Distrib (rOM_ClxFhorse, LogUniform,              0.00001,       0.3);
Distrib (f_Slope, TruncNormal_cv, 1, 1, 0, 10);
Distrib (dv, TruncNormal_cv, 1, 0.2, 0, 5);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter)

# ./mcsim.DEBcaliSWI DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	pAm.1.	Fm.1.	r_pAm_pM.1.	kap.1.	Ehp.1.	E_coc.1.
Result_mode	156.967000	0.79549600	0.10023400	0.89588400	177.2620	54.388400
2.5%	135.842200	0.60187945	0.10036700	0.85473285	149.3550	49.969300
97.5%	167.689000	0.79860100	0.13759655	0.89900000	212.5430	61.030100
Min	71.692400	0.05592130	0.10000100	0.31514900	103.4860	19.944000
Max	181.279000	0.79999800	0.86186600	0.89999800	247.6030	67.013900

Result_mean	150.905394	0.74624990	0.11233621	0.88245229	181.2955	55.400429
Result_sd	8.419143	0.05487286	0.01264795	0.01260473	15.9788	2.833521
	rOM_ClxFhorse.1.	f_Slope.1.	dv.1.	Sigma_W.1.		
Result_mode	5.288140e-04	0.6549470	1.9790300	0.18513100		
2.5%	7.540798e-05	0.5298000	1.7788600	0.16545900		
97.5%	3.857094e-03	1.9215895	2.2623135	0.22244300		
Min	1.018780e-05	0.4219580	0.7344160	0.14369800		
Max	1.902730e-02	2.9691600	2.6652800	5.43013000		
Result_mean	1.428443e-03	0.9296826	2.0027033	0.19281915		
Result_sd	1.012942e-03	0.3745798	0.1261005	0.04699618		

Number of observations, n = 277
 Number of parameters, k = 9
 Log-likelihood, LL = -79.05065
 BIC = 208.7175

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	1.02	1.05
E_coc.1.	1.00	1.00
Ehp.1.	1.04	1.12
f_Slope.1.	1.06	1.19
Fm.1.	1.04	1.10
kap.1.	1.01	1.03
pAm.1.	1.12	1.37
r_pAm_pM.1.	1.00	1.01
rOM_ClxFhorse.1.	1.04	1.14
Sigma_W.1.	1.05	1.15

Multivariate psrf

1.1

```

mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

```

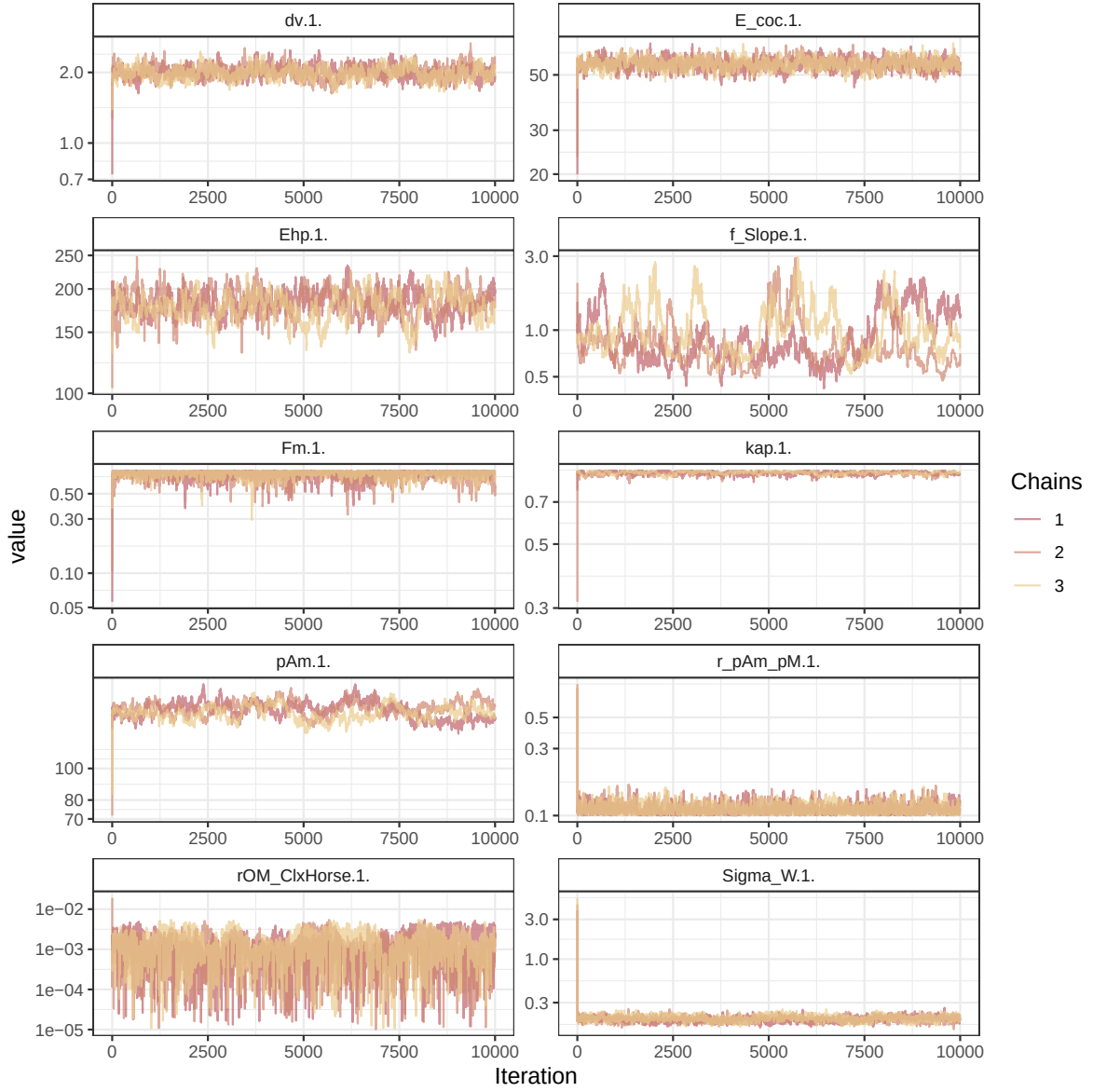


Figure 1: Parameters chains. Only the first iteration and the last r `Nb_iter_kept` are represented.

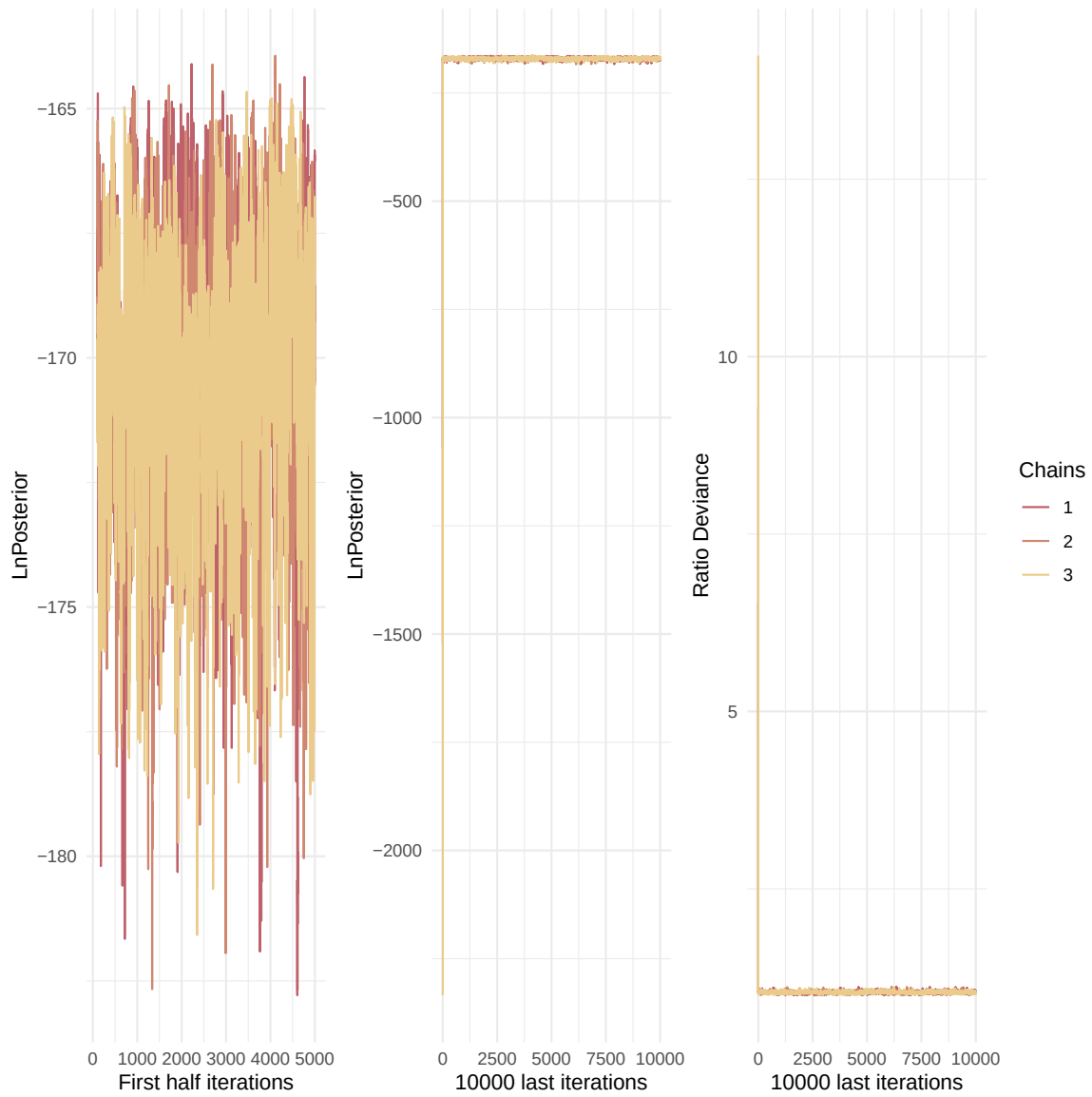


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

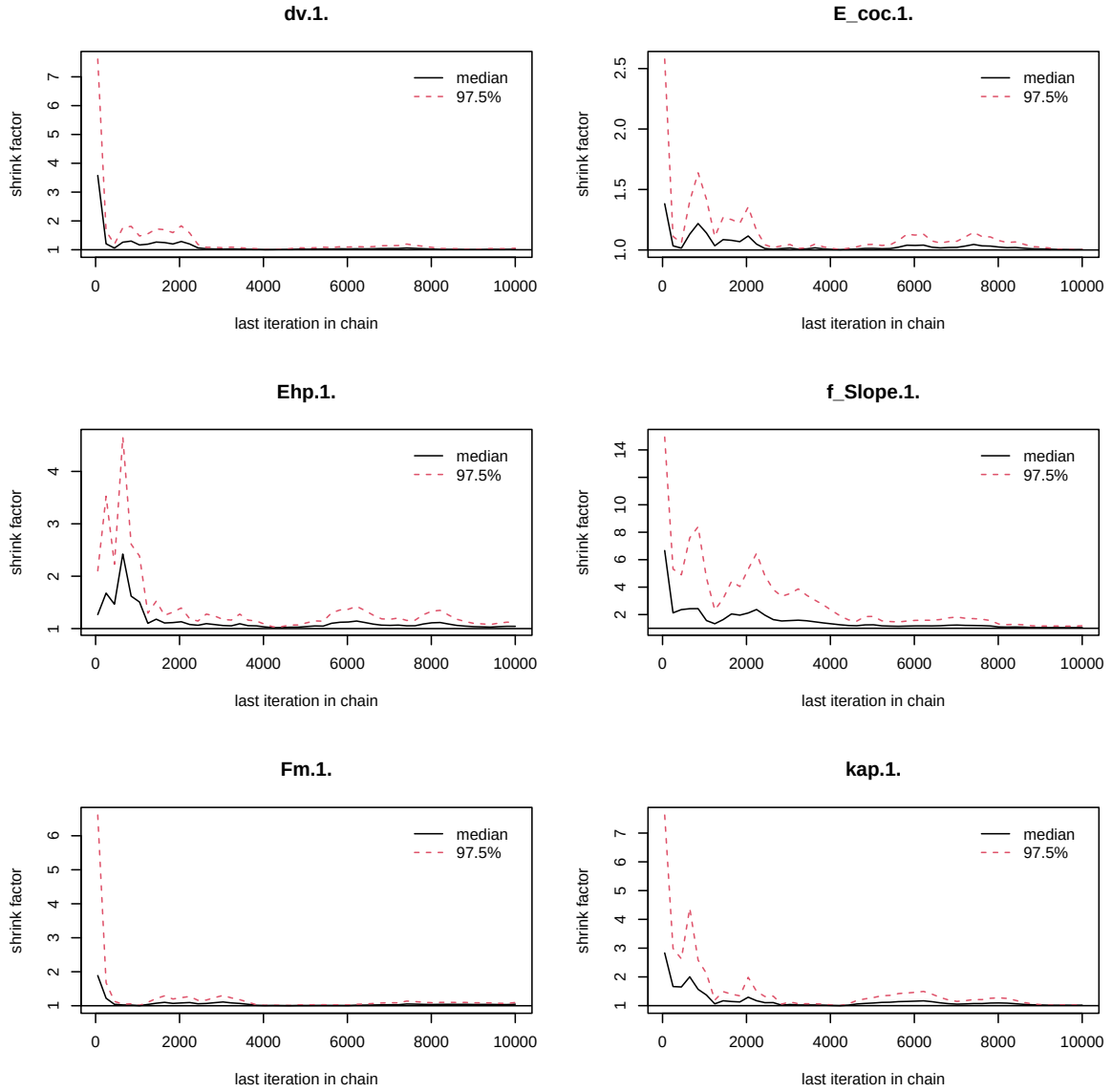


Figure 3: Chains convergence.

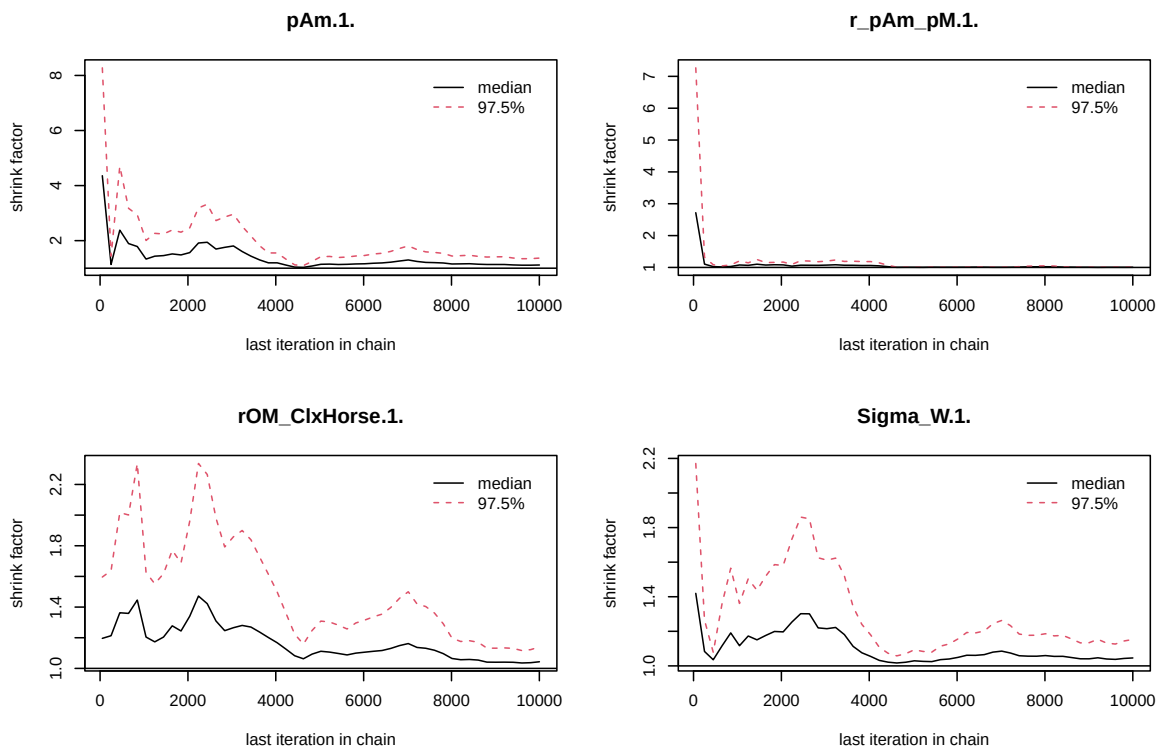


Figure 4: Chains convergence.

```
colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
pairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

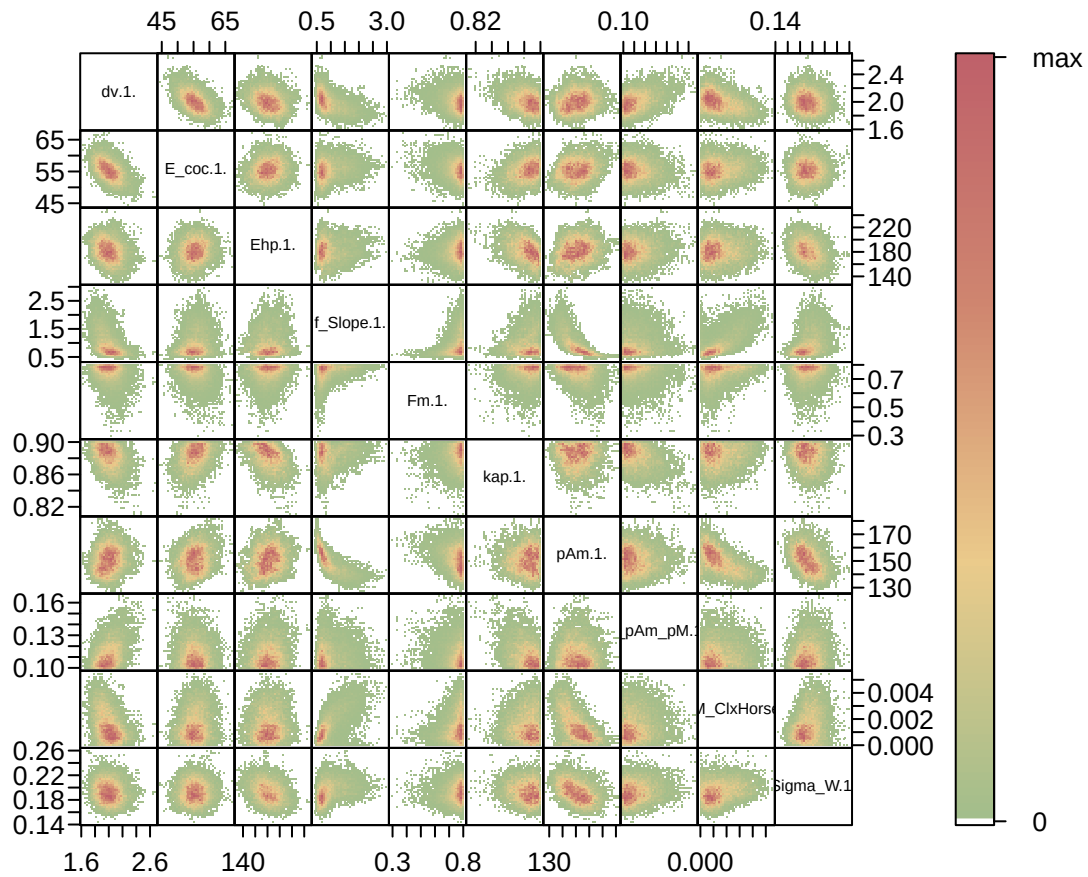


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
```

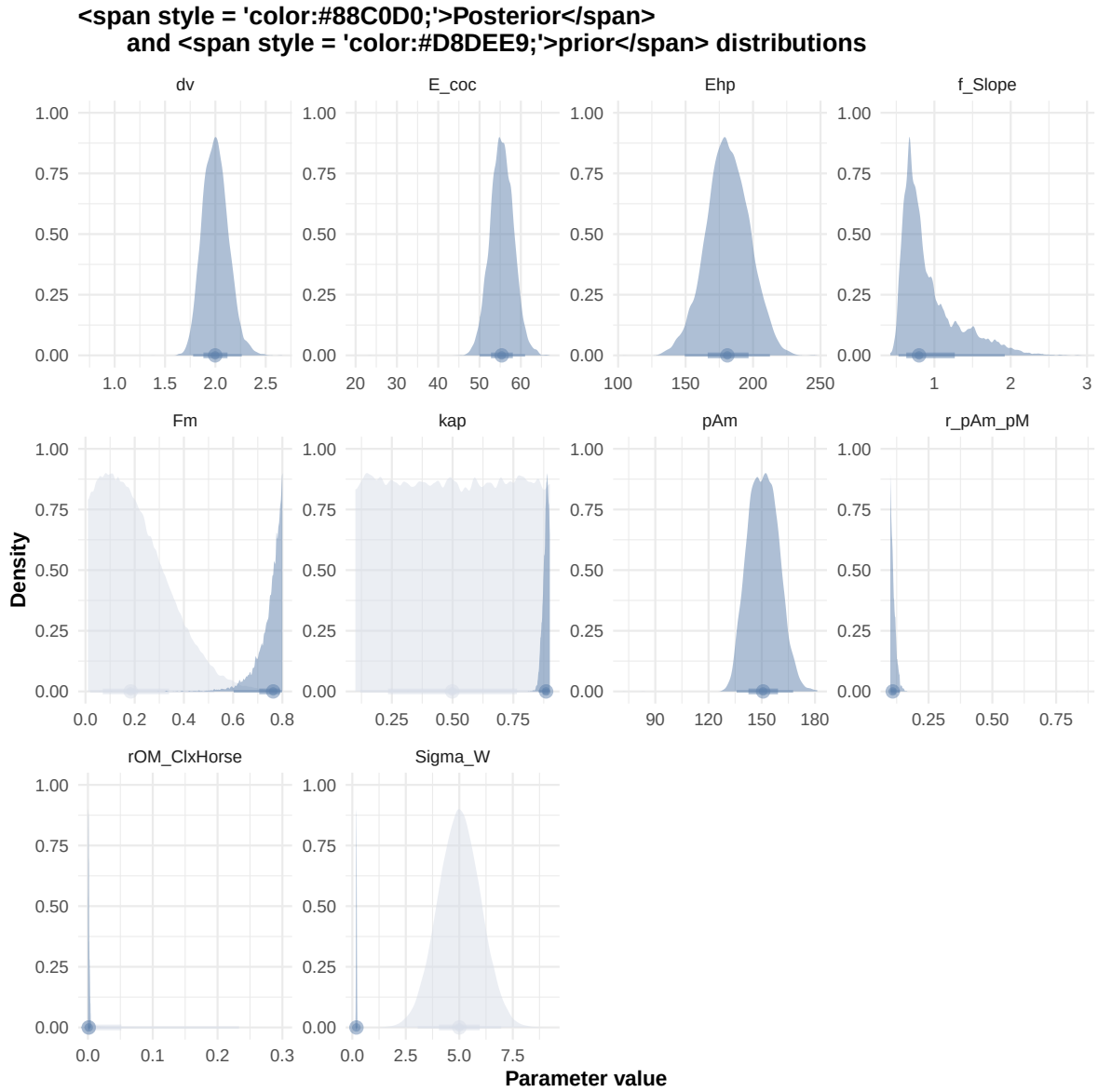


Figure 6: Distributions of the priors and the posteriors.

```

"Bart2019_XP3",
"Bart2019_XP4_1",
"Bart2019_XP4_2",
"Bart2019_XP5",
"Bart2020",
#"Gollot2026_lufa"
"Gollot2026_dens_D1",
"Gollot2026_dens_D2",
"Gollot2026_dens_D2b",
"Gollot2026_dens_D5",
"Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\.1\\.|$|\\.1\\. ", "", param),
    line = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
  Length_struct,
  Reproduction,
  Energy,
  Organic_matter,"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcaliSWI DEB_sim_full.in

#####
df_No_sim <- df_data |>

```

```

dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times)

```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected relationship between variables
 i Row 1 of `x` matches multiple rows in `y`.
 i Row 1 of `y` matches multiple rows in `x`.
 i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

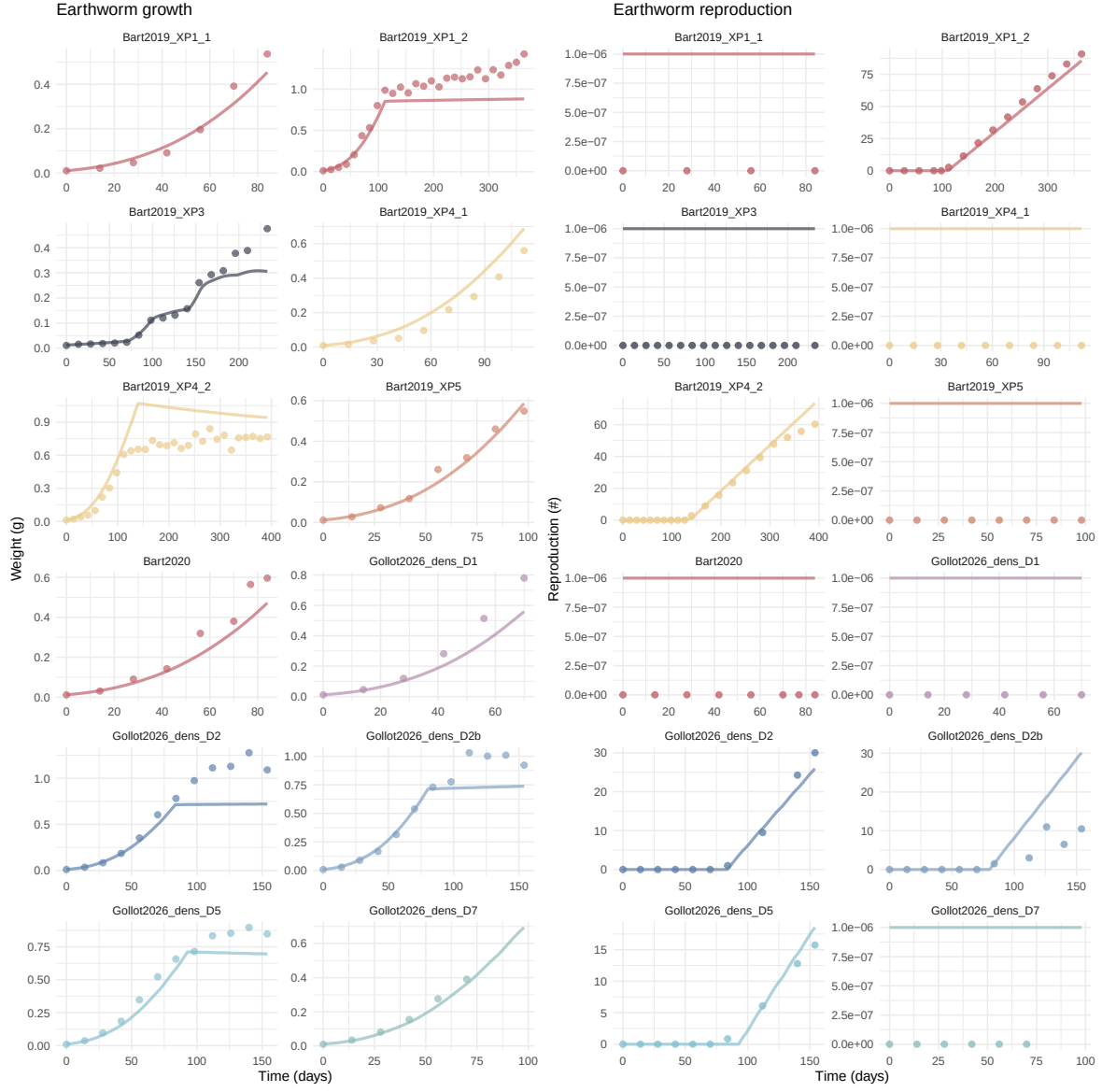
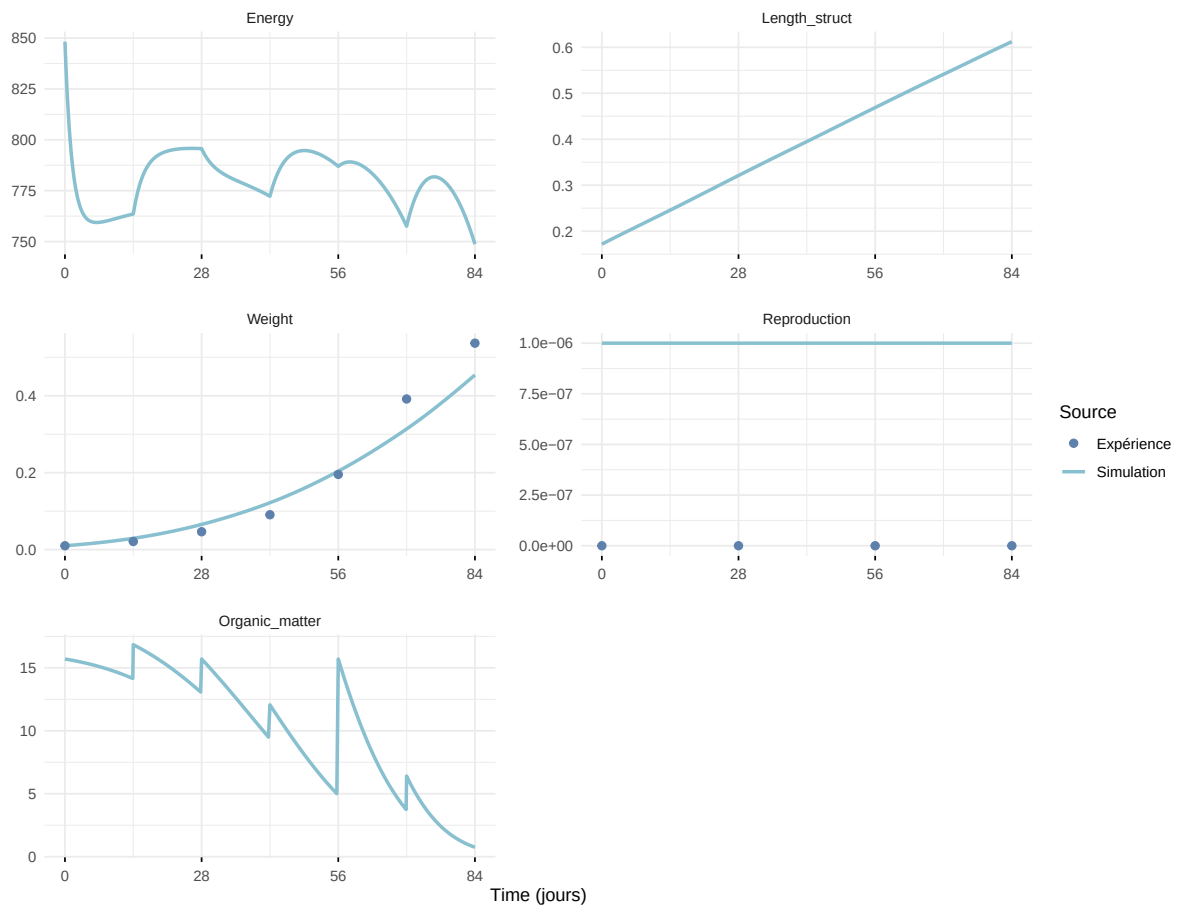
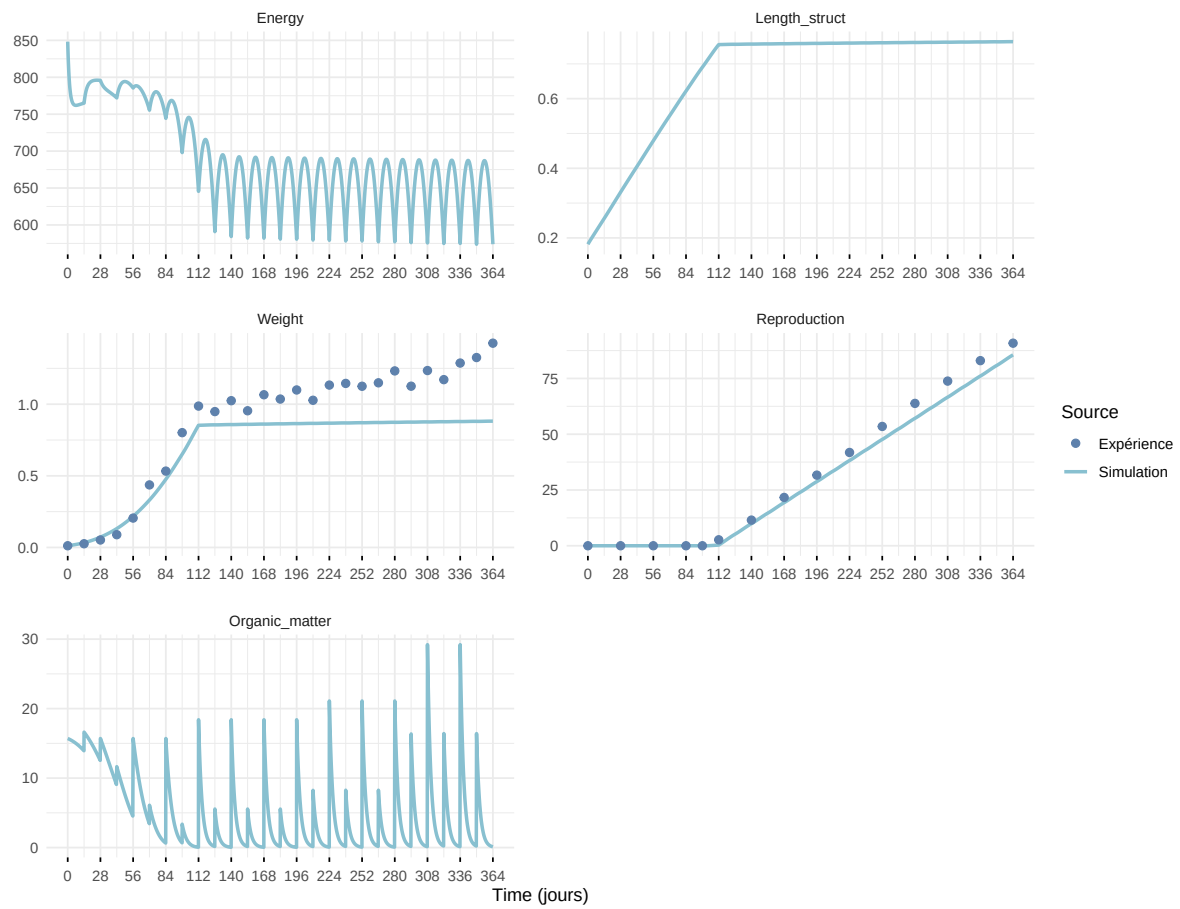


Figure 7: Simulations Weight and Reproduction.

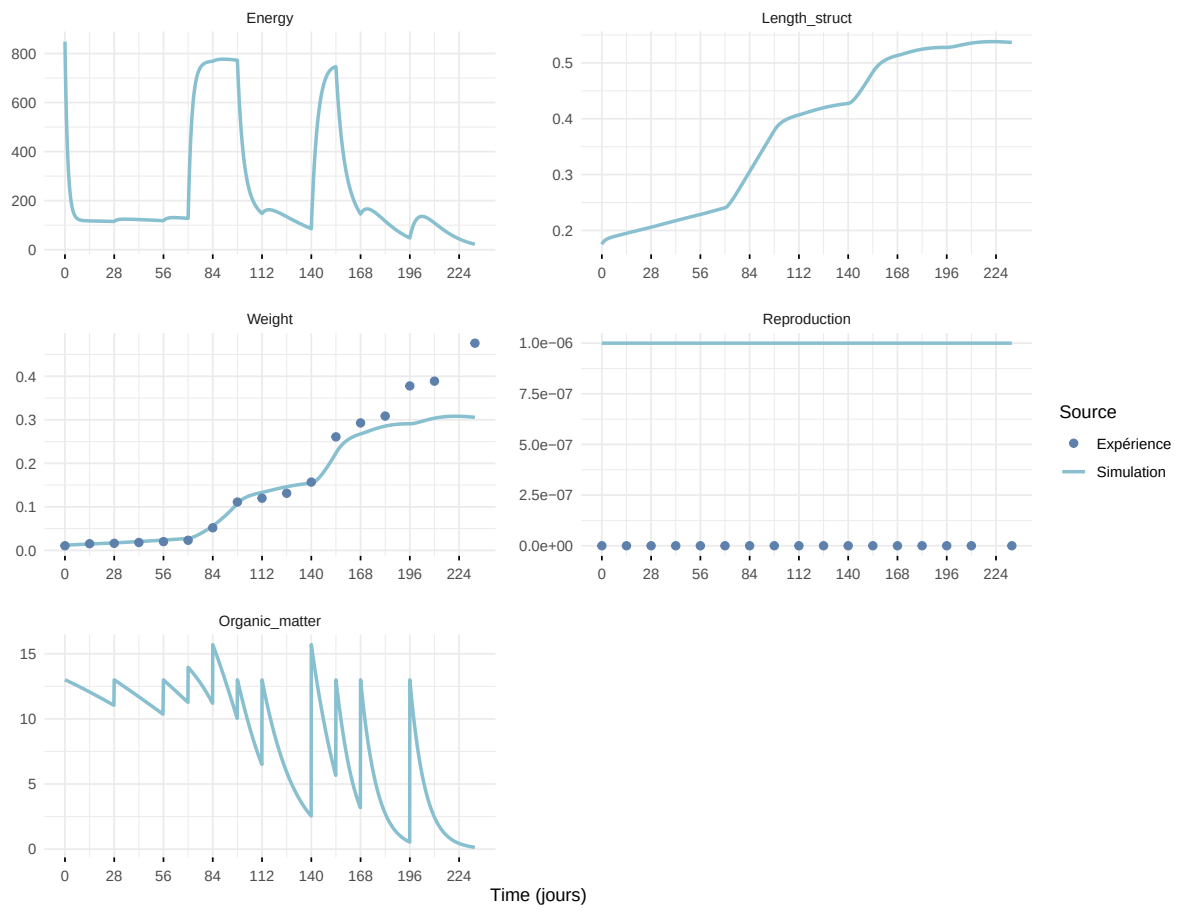
Expérience : Bart2019_XP1_1



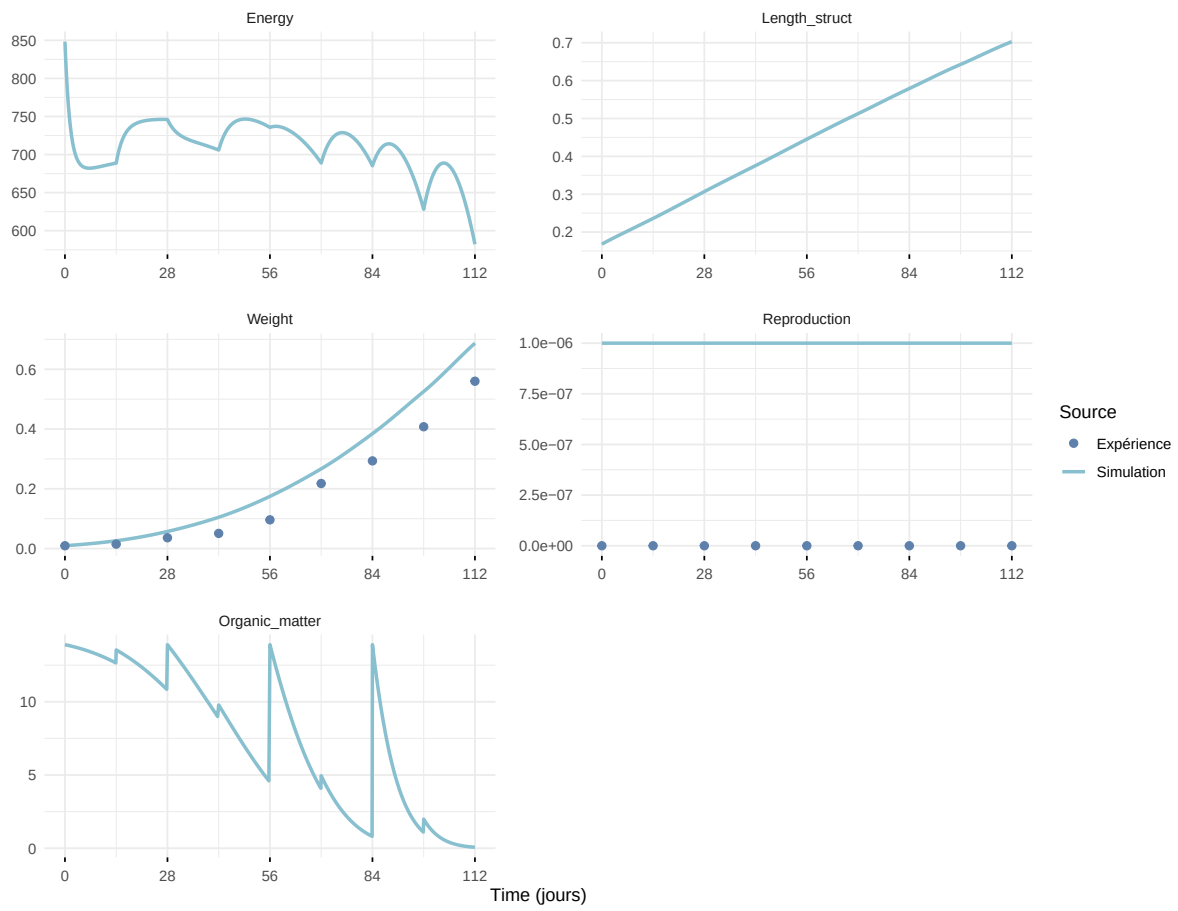
Expérience : Bart2019_XP1_2



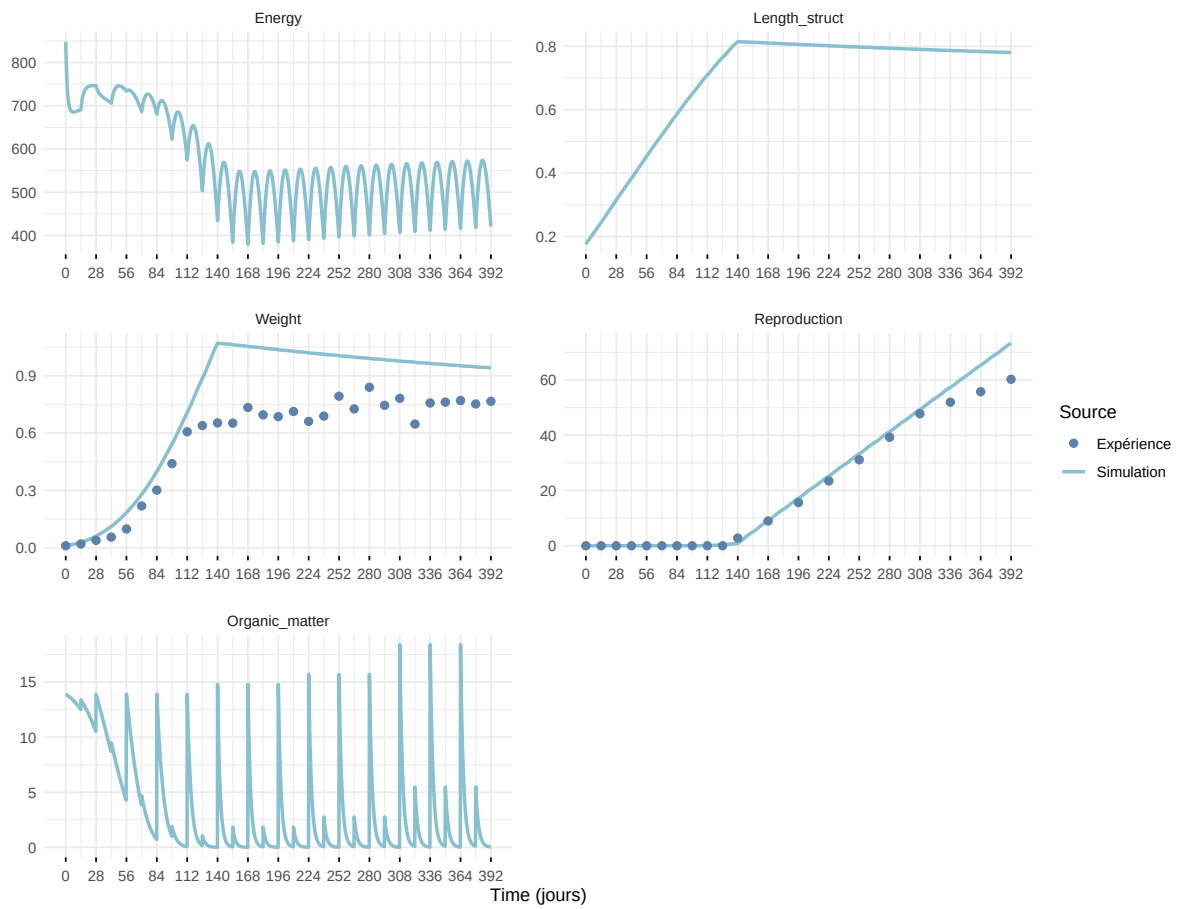
Expérience : Bart2019_XP3



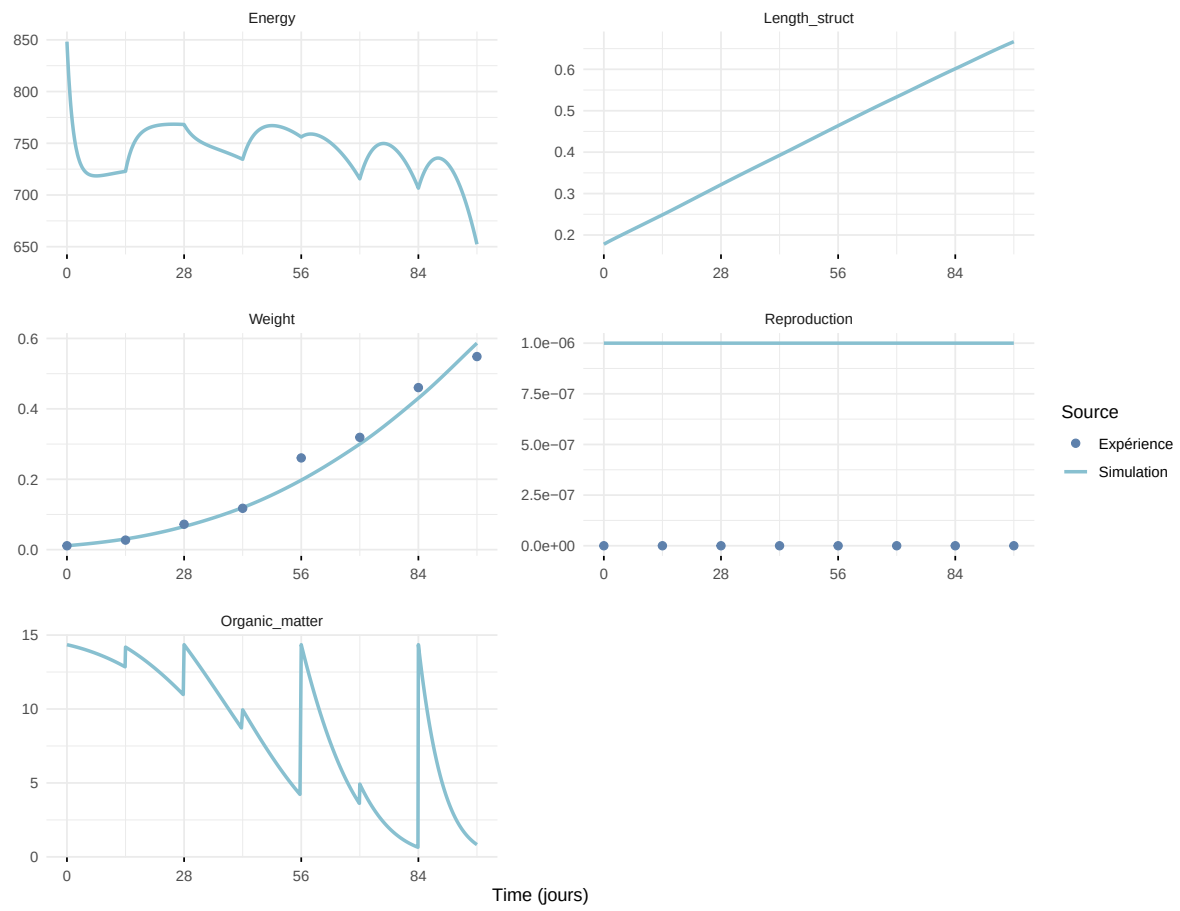
Expérience : Bart2019_XP4_1



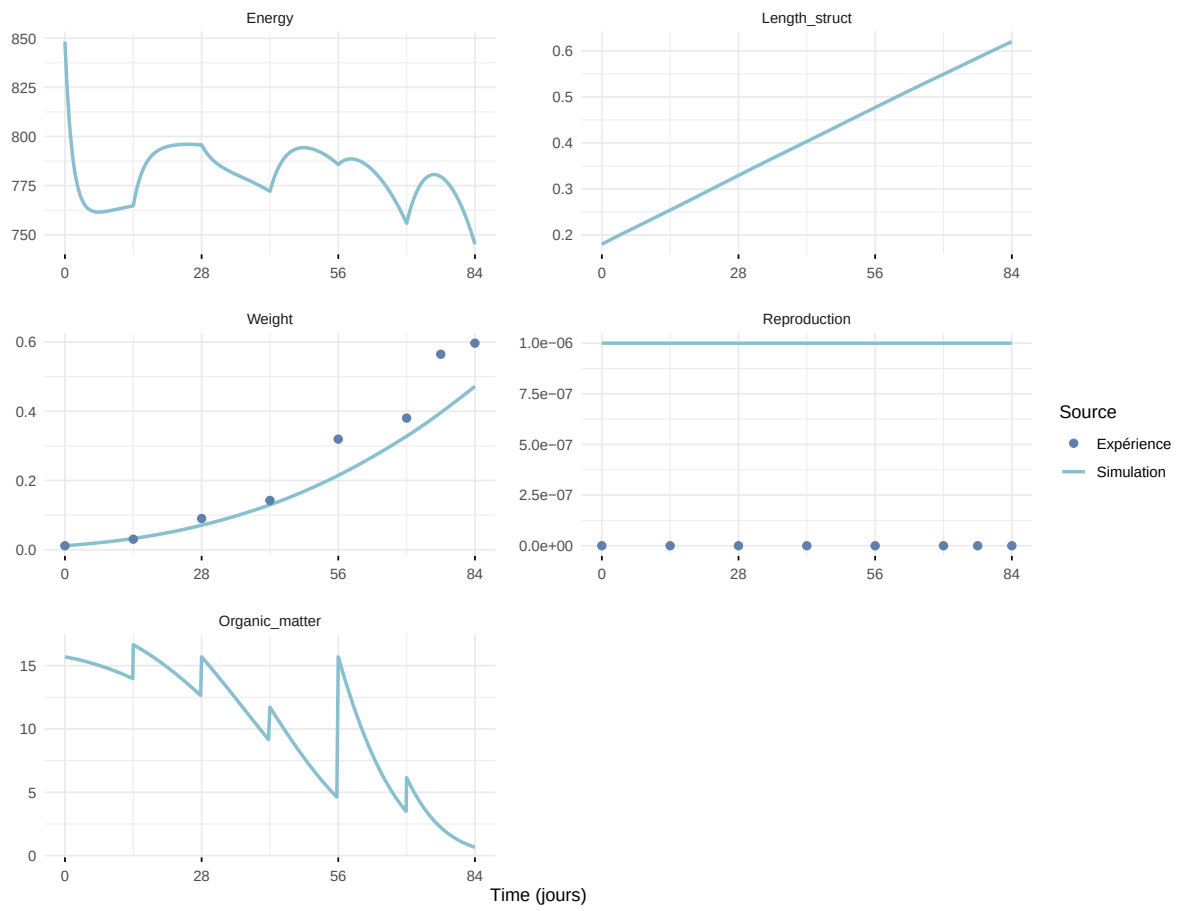
Expérience : Bart2019_XP4_2



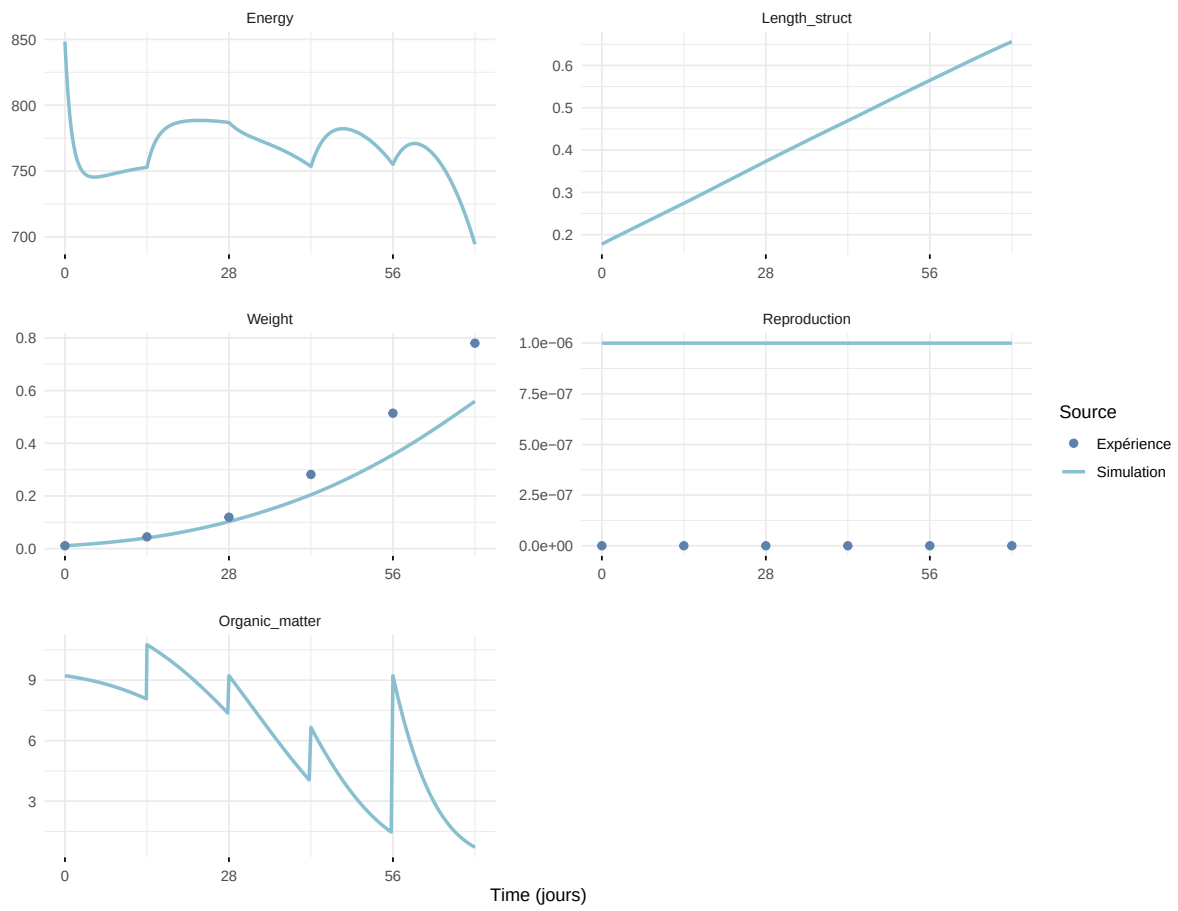
Expérience : Bart2019_XP5



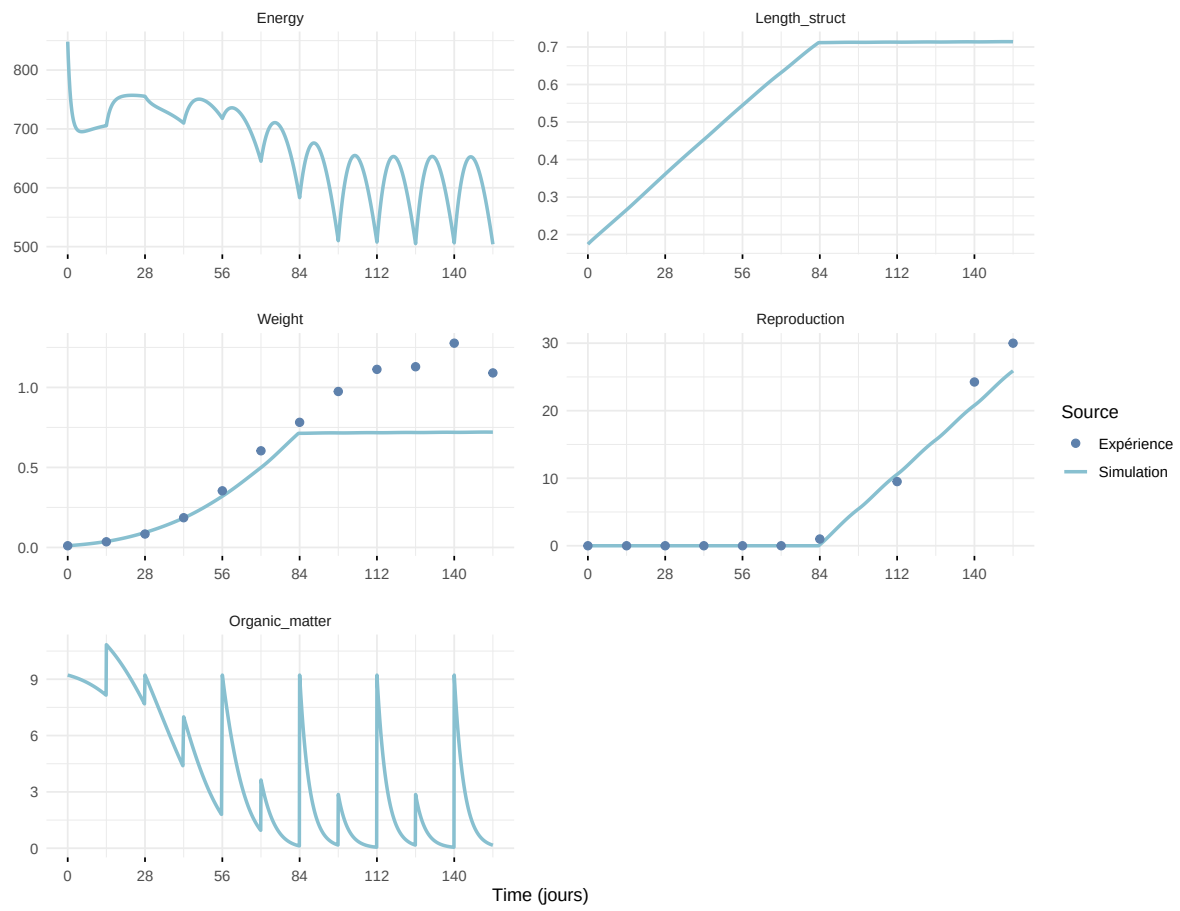
Expérience : Bart2020



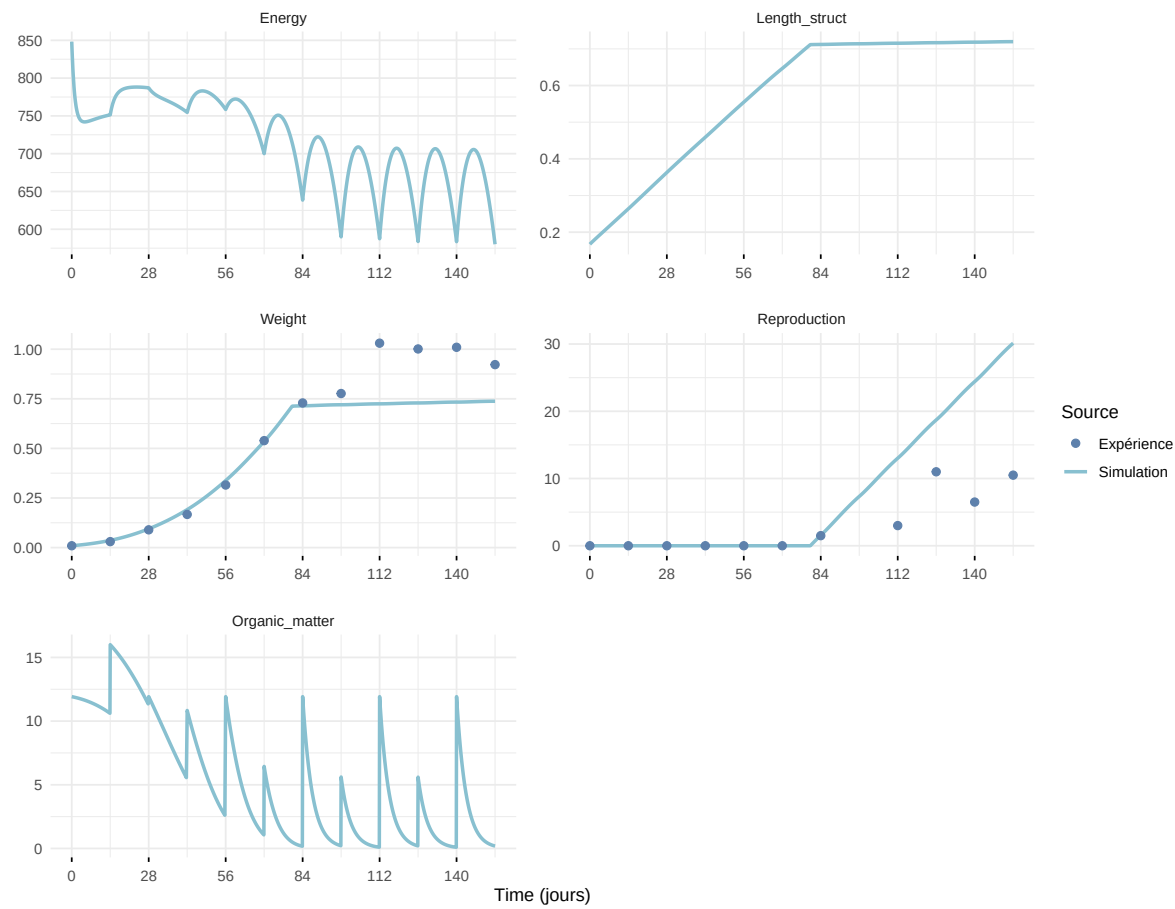
Expérience : Gollot2026_dens_D1



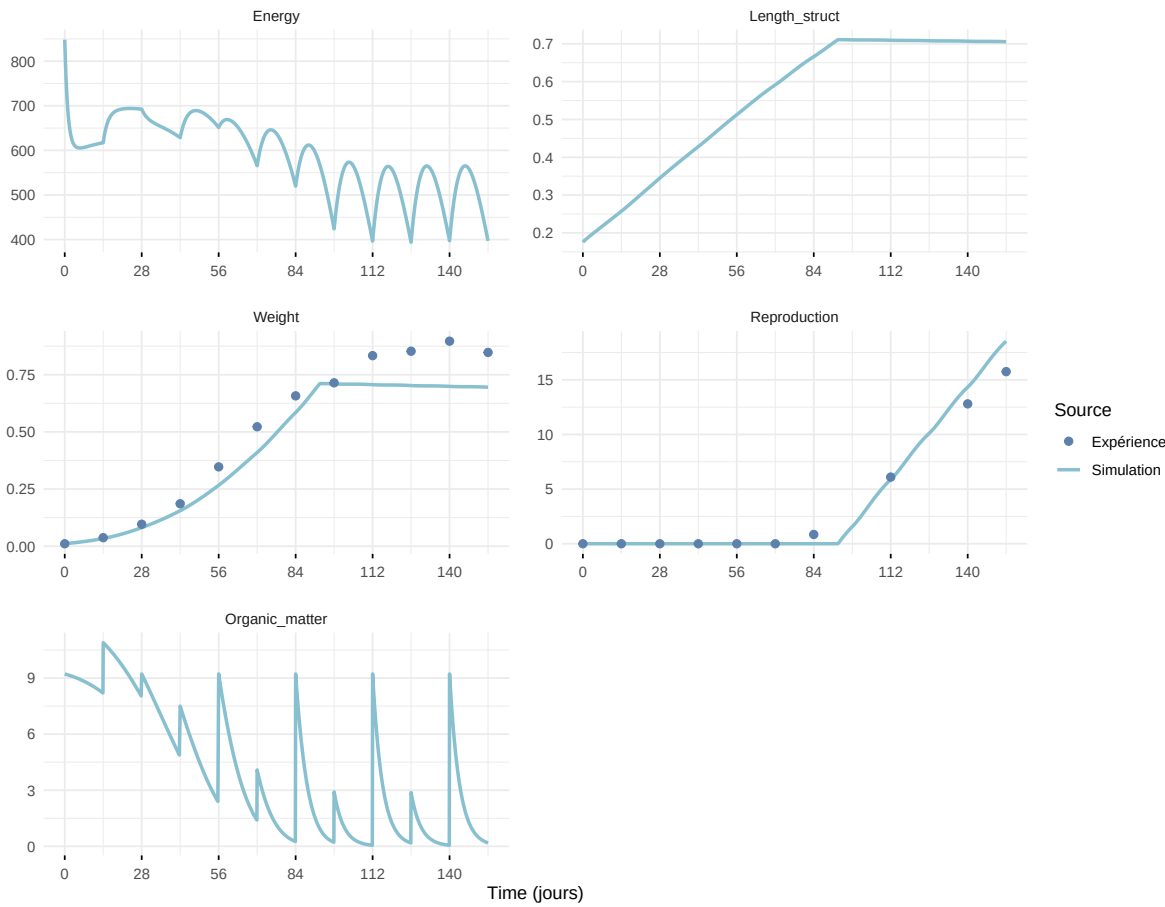
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

